

XIAOTIAN LIU

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 $\diamond$ Google Scholar: <https://scholar.google.com/citations?user=0CHV4aYAAAAJ>

EDUCATION

H. Milton Stewart School of Industrial and Systems Engineering, Georgia Institute of Technology *Atlanta, USA*
Ph.D. Candidate, Advised by Prof. Christos Alexopoulos and Prof. Edwin Romeijn *Sept. 2023 - 2027 (expected)*

School of Electronics Engineering and Computer Science, Peking University *Beijing, China*
B.S. in Electronics Engineering *Sept. 2018 - June 2022*

Guanghua School of Management, Peking University *Beijing, China*
B.S. in Management *Sept. 2019 - June 2022*

RESEARCH INTERESTS

My research focuses on pushing the frontier of operations management (OM) through deep neural networks, the key driver of the current AI wave. I investigate how to effectively train deep neural networks (DNNs) for modern OM problems involving high-dimensional data and nonlinear dynamics. I also work on combining DNNs with operational knowledge and exploiting advanced DNN architectures, such as transformers, for intelligent OM decision-making.

JOURNAL PUBLICATIONS

(^athesis advisor, ^bstudent collaborator, ^cprofessor collaborator, ^dthesis co-advisor)

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- [1] [Xiaotian Liu](#), Christos Alexopoulos^a, Yijie Peng^c, and Cheng Zhang^c, “Production Planning with Markovian Production Relationships,” *Production and Operations Management* (UTD24), 2026, <https://journals.sagepub.com/doi/10.1177/10591478261445695>
- [2] [Xiaotian Liu](#), Ming Hu^c, Yijie Peng^c, and Yaodong Yang^c, “Multi-Agent Deep Reinforcement Learning for Multi-Echelon Inventory Management,” *Production and Operations Management* (UTD24), 2025, <https://journals.sagepub.com/doi/full/10.1177/10591478241305863>
- *Production and Operations Management* 2023–2026 Top-20 Most Cited Paper (Journal Official Ranking)
 - The Only AI-Related Paper in The Top-20 Citation List
- [3] [Xiaotian Liu](#), Yijie Peng^c, Gongbo Zhang^b, and Ruihan Zhou^b, “An Efficient Node Selection Policy for Monte Carlo Tree Search with Neural Networks,” *INFORMS Journal on Computing* (UTD24), 2025, <https://pubsonline.informs.org/doi/full/10.1287/ijoc.2023.0307>
- *INFORMS Journal on Computing* Meritorious Paper Award (The Only Awardee in 2024)
 - The First Time This Award Was Presented to a Chinese Research Team
- [4] [Xiaotian Liu](#), Christos Alexopoulos^a, and Yijie Peng^c, “A Simulation-Driven Machine Learning Framework for Large-Scale Inventory Management,” *Annals of Operations Research* (ABS3), 2025, <https://link.springer.com/article/10.1007/s10479-025-06857-5>

WORKING PAPERS

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- [1] (Job Market Paper) Ming Hu^c, [Xiaotian Liu](#), and Guohao Shen^c (authors listed in alphabetical order), “AI as Flesh, OM as Bone: Semiparametric Regression with Deep Neural Networks,” Under revision (previous decision: reject and resubmit) at *Manufacturing & Service Operations Management* (UTD24), 2026, https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5460214
- [2] [Xiaotian Liu](#), Christos Alexopoulos^a, and Edwin Romeijn^d, “Learning Meets Convex Structure: Offline Approximation for Inventory Control,” Under review at *Production and Operations Management* (UTD24), 2026,

https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5393319

- Appeared in 2025 SSRN Top-10 Download List for Industrial & Manufacturing Engineering

[3] [Xiaotian Liu](#), Christos Alexopoulos^a, and Edwin Romeijn^d, “Adaptive Deployment of Black-Box Inventory Policies in Nonstationary Environments,” Work in progress, 2026

CONFERENCE PUBLICATIONS

[1] Qinghao Wang^b, Jinyang Jiang^b, [Xiaotian Liu](#), Tao Ren^b, Yi Zheng^b, Cheng Zhang^c, Yaodong Yang^c, and Yijie Peng^c, “A Unified Framework for Multi-Stage Decision Optimization with Deep Reinforcement Learning and Foundation Models,” *2025 IEEE 21st International Conference on Automation Science and Engineering (CASE)*, 2025, <https://ieeexplore.ieee.org/abstract/document/11163883>

[2] Gongbo Zhang^b, [Xiaotian Liu](#), and Yijie Peng^c, “A Simulation Optimization Method for Scheduling Automated Guided Vehicles in a Stochastic Warehouse Management System,” *2023 Winter Simulation Conference (WSC)*, 2023, <https://ieeexplore.ieee.org/abstract/document/10407793>

- Second Place Winning Method in *2022 Winter Simulation Conference Case Study Competition*

[3] [Xiaotian Liu](#), Lei Yang^b, Xiaoyu Zhang^b, and Xiaohui Duan^c, “MA-ResNet50: A General Encoder Network for Video Segmentation,” *2022 International Conference on Computer Vision Theory and Applications (VISAPP)*, 2022, <https://pdfs.semanticscholar.org/3d6f/8f0a026029521815c6402f6c52f937fc7196.pdf>

PATENTS

[1] Yijie Peng, [Xiaotian Liu](#), Gongbo Zhang, Cheng Zhang, and Min Zhan, “An Adaptive Bitrate Control Method and System,” Chinese Invention Patent No. ZL202510949984.6, 2025.

HONORS & AWARDS

INFORMS Journal on Computing Meritorious Paper	2025
Second Place Award (ranking 2/31), Winter Simulation Conference 2022 Case Study Competition	2022
Mao Yugang Scholarship for Outstanding Undergraduate in Research (Top 1%, Peking University)	2022
Outstanding Undergraduate Research Program (Top 1%, Peking University)	2021

TEACHING

Deterministic Optimization (ISYE6669) <i>Teaching Assistant</i>	<i>Georgia Institute of Technology, USA</i> <i>Spring 2025</i>
Computational Data Analysis (ISYE6740) <i>Teaching Assistant</i>	<i>Georgia Institute of Technology, USA</i> <i>Summer 2024, Fall 2024, Summer 2025</i>

PRESENTATIONS / INVITED TALKS

A Unified Framework for Decision Optimization with Deep Reinforcement Learning and Foundation Models
2025 International Conference on Automation Science and Engineering *Los Angeles, California, USA, 2025*

A Simulation-Driven Machine Learning Framework for Large-Scale Inventory Management
2023 INFORMS Annual Meeting *Phoenix, Arizona, USA, 2023*

Second Place Winner Presentation for WSC Data Competition
2022 Winter Simulation Conference *Singapore, 2022*

MA-ResNet50: A General Encoder Network for Video Segmentation
2022 International Conference on Computer Vision Theory and Applications *Online, 2022*

ACADEMIC SERVICE

Paper Reviewer: INFORMS Journal on Computing; Winter Simulation Conference